



OFFLINE-FIRST SECURITY ENGINEERING

VNAGY Coding Standard Specification

Internal Technical Standard for Minimal, Deterministic, and Explicit Code Practices

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Abstract

This document defines the official VNAGY coding standard for minimal, deterministic, and explicit software development practices across all VNAGY Security Engineering projects. It establishes mandatory rules for code structure, naming, formatting, documentation, and architectural consistency to ensure reliability, offline-first resilience, and long-term maintainability. The specification provides a unified baseline for all contributors, enabling predictable behavior, reduced ambiguity, and strict alignment with VNAGY's security-driven engineering principles.

Scope

This specification applies to all **VNAGY Security Engineering software projects**, modules, libraries, and internal tools. It defines mandatory coding rules, formatting requirements, naming conventions, documentation standards, and structural guidelines that must be followed by all contributors. The scope includes source code, configuration files, internal scripts, and supporting development assets. The specification does not cover product design, UI/UX guidelines, infrastructure deployment, or operational security procedures.

VNAGY

Definitions

Deterministic Code

Code that produces the same output for the same input without hidden or implicit state. Determinism ensures predictability, testability, and offline-first reliability.

Explicit State

State that is fully visible, controlled, documented, and intentionally managed. No hidden transitions or implicit mutations are allowed.

Minimal Surface

The smallest possible public API or module interface required for functionality. Reduces complexity, attack surface, and maintenance overhead.

Offline-First

A design principle where software must operate without network connectivity, ensuring resilience, determinism, and security regardless of external dependencies.

Module

A self-contained, deterministic unit of functionality with explicit inputs, outputs, and documentation. Modules must follow **VNAGY** structural and naming rules.

Component

A higher-level construct composed of multiple modules. Components must maintain deterministic behavior and minimal coupling.

Contributor

Any individual who writes, modifies, or maintains **VNAGY** code, documentation, or development assets.

Internal Tool

A **VNAGY**-maintained script, utility, or automation asset used to support development, testing, or deployment workflows.

Coding Standard

A mandatory set of rules governing code structure, naming, formatting, documentation, and deterministic behavior across all **VNAGY** projects.

Security Engineering

The discipline focused on designing and implementing systems that maintain predictable, secure behavior under all conditions, including offline operation.

Configuration Artifact

Any file or asset that defines system behavior, module parameters, or environment settings. Must follow explicit formatting and documentation rules.

Development Asset

Supporting files used during development, such as scripts, templates, or metadata. Must adhere to **VNAGY** structural and naming conventions.

Normative References

The following documents are referenced as authoritative sources and form the normative foundation for this specification:

- **VNAGY Architecture Specification** — defines structural, modular, and offline-first design principles used across all VNAGY systems.
- **VNAGY Security Principles** — establishes mandatory security requirements and deterministic behavior rules.
- **VNAGY Offline-First Model** — outlines constraints and guarantees for systems operating without network dependency.
- **ISO/IEC 27001** — international standard for information security management systems.
- **NIST SP 800-63** — digital identity guidelines providing structure for deterministic and secure authentication flows.
- **Microsoft SDL Coding Guidelines** — secure development lifecycle rules influencing VNAGY coding and documentation practices.

Compliance Requirements

The following requirements define mandatory and recommended rules that all contributors must follow when writing, modifying, or maintaining **VNAGY** software components. These compliance statements ensure deterministic behavior, explicit state management, minimalism, and offline-first reliability across all **VNAGY** systems.

Mandatory Requirements

Code **MUST** be deterministic and produce identical results for identical inputs.

- All state **MUST** be explicit, documented, and intentionally controlled.
- Modules **MUST** follow **VNAGY** structural and naming conventions.
- Public interfaces **MUST** expose a minimal surface.
- Error handling **MUST** be explicit, predictable, and non-implicit.
- All configuration artifacts **MUST** follow **VNAGY** formatting and documentation rules.
- All code **MUST** be compatible with offline-first operation.
- Security-relevant logic **MUST** avoid dynamic or implicit behavior.

Prohibited Requirements

- Code **MUST NOT** introduce hidden state or implicit mutations.
- Modules **MUST NOT** depend on unpredictable external systems (e.g., network availability).
- Components **MUST NOT** expose unnecessary public functions or data.
- Error handling **MUST NOT** rely on silent failures or implicit fallbacks.
- Contributors **MUST NOT** bypass **VNAGY** documentation or formatting rules.

Recommended Requirements

- Modules **SHOULD** minimize dependencies and coupling.
- Components **SHOULD** maintain clear separation of concerns.

- Documentation **SHOULD** be concise, minimal, and placed close to the code it describes.
- Contributors **SHOULD** use **VNAGY** internal tools for validation, formatting, and consistency checks.

Optional Requirements

- Components **MAY** extend functionality if extensions remain deterministic and minimal.
- Modules **MAY** include internal helper functions if they do not expand the public surface.
- Contributors **MAY** propose new coding rules if they align with **VNAGY** principles.

Coding Principles

Minimalism

Code must expose the smallest possible interface, contain only essential logic, and avoid unnecessary abstractions. Minimalism reduces complexity, attack surface, and maintenance overhead.

Determinism

All code must behave predictably and produce identical results for identical inputs. Deterministic execution is mandatory for offline-first reliability, testing, and security guarantees.

Explicitness

State, behavior, dependencies, and data flows must be fully visible and intentionally controlled. No implicit mutations, hidden transitions, or ambiguous logic are allowed.

Isolation

Modules must be self-contained, independent, and free of unnecessary coupling. Each module must define clear inputs, outputs, and responsibilities.

Predictability

Code must avoid dynamic behavior, runtime surprises, and implicit fallbacks. Predictable execution ensures stable operation in constrained or offline environments.

Consistency

Naming, formatting, structure, and documentation must follow **VNAGY** standards uniformly across all projects. Consistency enables readability, maintainability, and cross-team collaboration.

Offline-First Reliability

All code must operate correctly without network access. External dependencies must be optional, controlled, and never required for core functionality.

Security by Design

Code must follow safe defaults, avoid unnecessary exposure, and minimize trust assumptions. Security is embedded into every module, not added later.

VNAGY

Code Structure

VNAGY codebases must follow a predictable, minimal, and deterministic structure. All modules, components, and development assets must be organized in a way that ensures clarity, isolation, and offline-first reliability. The following structural rules apply to all **VNAGY** software projects.

Project Root Layout

Every **VNAGY** project **MUST** follow the standard root layout:

- **src** — contains all production modules and components
- **config** — explicit configuration artifacts
- **docs** — internal documentation, specifications, and module descriptions
- **tests** — deterministic test suites
- **integration** — integration logic, adapters, or bridges
- **tools** — internal scripts and development assets
- **profiles** — environment-specific offline-first profiles (optional)

This structure ensures consistency across all **VNAGY** repositories.

Module Structure

Each module **MUST** be self-contained and follow this internal layout:

- **module_name/**
 - **main.vng** — primary deterministic logic
 - **state.vng** — explicit state definitions
 - **errors.vng** — explicit error types
 - **types.vng** — data structures and interfaces
 - **README.md** — minimal documentation
 - **tests/** — module-specific deterministic tests

Modules **MUST NOT** expose unnecessary files or dynamic behavior.

Component Structure

Components are composed of multiple modules and **SHOULD** follow this pattern:

- **component_name/**
 - **modules/** — isolated modules
 - **adapter/** — deterministic interfaces to external systems
 - **offline/** — offline-first logic and fallbacks
 - **config/** — component-specific configuration artifacts
 - **README.md**

Components **MUST** maintain minimal coupling and explicit boundaries.

File Naming Rules

- File names **MUST** be lowercase with hyphens: `module-name.vng`
- Directories **MUST** use lowercase: `module_name/`
- No spaces, no camelCase, no dynamic naming
- Public files **MUST** reflect their purpose:
 - `errors.vng`
 - `state.vng`
 - `types.vng`
 - `main.vng`

This ensures predictability and readability.

Deterministic Folder Behavior

- No folder may contain mixed responsibilities
- No folder may contain hidden state
- No folder may contain dynamic runtime artifacts
- All folders **MUST** be safe for offline operation

Separation of Concerns

- Logic, state, errors, and types **MUST** be separated
- Documentation **MUST** be placed next to the code it describes
- Tests **MUST** mirror the structure of the modules they validate

Prohibited Structures

The following patterns **MUST NOT** appear in **VNAGY** projects:

- `utils/` folders with mixed logic
- `helpers/` folders without explicit purpose
- deeply nested dynamic structures
- implicit module grouping
- runtime-generated directories

VNAGY architecture requires strict determinism and explicitness.

Naming Conventions

All names used in **VNAGY** software components **MUST** follow strict deterministic and minimal rules. Naming conventions ensure clarity, predictability, and consistency across all **VNAGY** modules, components, and configuration artifacts.

Directory Naming

- Directories **MUST** use lowercase.
- Words **MUST** be separated with hyphens or underscores.
- Directory names **MUST** reflect their purpose.

Examples:

- **src/**
- **config/**
- **tests/**
- **module-name/**
- **component_name/**

Module Naming

- Module names **MUST** be lowercase.
- Module names **MUST** be short and descriptive.
- Module names **MUST NOT** contain spaces or camelCase.
- Module names **SHOULD** reflect deterministic functionality.

Examples:

- **auth-module/**
- **device-detection/**
- **profile-loader/**

Component Naming

- Components **MUST** use lowercase.
- Components **MUST** describe the system-level responsibility.
- Components **SHOULD** avoid abbreviations unless widely understood.

Examples:

- `agent/`
- `agent-installer/`
- `integration-layer/`

Type Naming

- Types **MUST** use PascalCase.
- Types **MUST** be explicit and descriptive.
- Types **MUST NOT** use abbreviations.

Examples:

- `UserProfile`
- `DeviceState`
- `ErrorCode`
- `OfflineContext`

Error Naming

- Error identifiers **MUST** use PascalCase.
- Error names **MUST** end with Error.
- Error names **MUST** describe the failure explicitly.

Examples:

- **InvalidStateError**
- **MissingConfigError**
- **OfflineModeError**

Constant Naming

- Constants **MUST** use UPPERCASE.
- Words **MUST** be separated with underscores.
- Constant names **MUST** be deterministic and explicit.

Examples:

- **DEFAULT_TIMEOUT**
- **MAX_RETRY_COUNT**
- **OFFLINE_MODE_ENABLED**

Function Naming

- Functions **MUST** use lowercase with hyphens or underscores.
- Function names **MUST** describe deterministic behavior.
- Function names **MUST NOT** be ambiguous.

Examples:

- **load-profile()**
- **validate-state()**
- **resolve-errors()**
- **apply-config()**

Variable Naming

- Variables **MUST** use lowercase with hyphens or underscores.
- Variable names **MUST** be minimal but descriptive.
- Variables **MUST NOT** use camelCase.

Examples:

- `current_state`
- `device_id`
- `profile_path`

Prohibited Naming Patterns

The following patterns **MUST NOT** appear in VNAGY codebases:

- camelCase (`deviceId`, `userProfile`)
- ambiguous names (`temp`, `data`, `info`)
- dynamic names (`file1`, `file2`, `moduleX`)
- mixed-case directories (`ModuleName/`)
- names without purpose (`misc/`, `helpers/`, `utils/`)

Formatting Rules

All **VNAGY** documents **MUST** follow strict formatting rules to ensure clarity, consistency, and deterministic presentation across all specifications, modules, and whitepapers. Formatting is not aesthetic; it is a structural requirement.

Typography Rules

- Section titles (**H1**) **MUST** use **24 pt**, bold.
- Subsection titles (**H2**) **MUST** use **16 pt**, bold.
- Third-level titles (**H3**) **MUST** use **12 pt**, bold.
- Body text **MUST** use **11 pt** regular.
- Notes, examples, and footnotes **MAY** use **10 pt**.

Alignment Rules

- All text **MUST** be left-aligned.
- Titles **MUST NOT** be centered.
- Code examples **MUST** use monospaced alignment.
- Tables **MUST** be left-aligned with consistent column width.

Color and Highlighting

- VNAGY documents **MUST** use monochrome formatting (black/gray).
- Colors **MUST NOT** be used for emphasis.
- Bold **MAY** be used for key terms.
- Italics **MAY** be used for definitions or references.

Lists and Structure

- Ordered lists **MUST** use Arabic numerals (1, 2, 3).
- Unordered lists **MUST** use hyphens (-).
- Nested lists **SHOULD NOT** be used.
- Each list item **MUST** be a complete, meaningful statement.

Tables

- Tables **MUST** have a header row.
- Columns **MUST** be left-aligned.
- Tables **MUST NOT** contain merged cells.
- Tables **MUST** be used only for structured comparisons or definitions.

Code Formatting

- Code blocks **MUST** use monospaced font.
- Code examples **MUST** be minimal and deterministic.
- Code **MUST NOT** include dynamic or ambiguous placeholders.
- File names **MUST** be shown exactly as used (`main.vng`, `state.vng`).

Document Layout

- Margins **MUST** use asymmetric layout:
 - **Left: 2.5 cm**
 - **Right: 2.0 cm**
 - **Top: 2.5 cm**
 - **Bottom: 2.0 cm**
- Page numbers **MUST** be bottom-center or bottom-right.
- Headers **MAY** include document version, section title, or classification.
- Footers **MUST NOT** contain decorative elements or non-functional graphics.

Prohibited Formatting

The following formatting patterns **MUST NOT** appear in VNAGY documents:

- centered titles
- colored text
- decorative fonts
- inconsistent spacing
- mixed alignment
- camelCase in headings
- oversized or undersized typography
- nested lists deeper than one level

Documentation Rules

All **VNAGY** documentation **MUST** follow strict deterministic, minimal, and explicit rules. Documentation is not decoration; it is a functional component of the **VNAGY** architecture.

General Documentation Principles

- Documentation **MUST** be minimal, explicit, and deterministic.
- Documentation **MUST NOT** contain narrative, filler, or subjective language.
- Documentation **MUST** describe behavior, structure, and constraints.
- Documentation **MUST** be updated whenever code structure changes.
- Documentation **MUST** be stored alongside the code it describes.

Required Documentation Types

VNAGY requires the following documentation artifacts:

- **Module Documentation** — describes module purpose, inputs, outputs, constraints.
- **Component Documentation** — describes component boundaries and responsibilities.
- **State Documentation** — describes deterministic state transitions.
- **Error Documentation** — describes explicit error types and conditions.
- **Configuration Documentation** — describes all configurable parameters.
- **README.md** — minimal overview of the directory or module.

Module Documentation Rules

Each module **MUST** include a `README.md` containing:

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- **Purpose** — one sentence.
- **Inputs** — explicit list.
- **Outputs** — explicit list.
- **State Dependencies** — explicit list.
- **Error Conditions** — explicit list.
- **Deterministic Guarantees** — explicit list.

Module documentation **MUST NOT** include:

- narrative explanations
- implementation details
- implicit behavior
- dynamic or runtime-dependent descriptions

State Documentation Rules

State documentation **MUST** include:

- **State Definitions**
- **Allowed Transitions**
- **Transition Preconditions**
- **Transition Postconditions**
- **Error States**

State documentation **MUST NOT** include:

- implicit transitions
- dynamic state mutation
- undocumented fallback behavior

Error Documentation Rules

Error documentation **MUST** include:

- **Error Name**
- **Error Condition**
- **Error Severity**
- **Error Recovery Path**
- **Error Determinism Guarantee**

Error documentation **MUST NOT** include:

- ambiguous error descriptions
- catch-all error categories
- implicit error behavior

Configuration Documentation Rules

Configuration documentation **MUST** include:

- **Parameter Name**
- **Parameter Type**
- **Default Value**
- **Allowed Values**
- **Deterministic Effect**
- **Offline-First Behavior**

Configuration documentation **MUST NOT** include:

- hidden parameters
- dynamic defaults
- undocumented side effects

README Rules

Every directory **MUST** contain a README.md with:

- **Purpose**
- **Contents**
- **Dependencies**
- **Constraints**

README **MUST NOT** contain:

- narrative text
- implementation details
- personal notes

Prohibited Documentation Patterns

The following patterns **MUST NOT** appear in VNAGY documentation:

- narrative storytelling
- subjective language
- ambiguous diagrams
- camelCase in headings
- undocumented behavior
- implicit assumptions
- dynamic descriptions
- "TODO" placeholders
- mixed formatting styles
- inconsistent terminology

VNAGY Security Requirements

All **VNAGY** components **MUST** follow strict security rules ensuring deterministic, isolated, offline-first, and non-reconstructable behavior. Security is not a feature; it is a structural constraint of the **VNAGY** architecture.

Isolation Requirements

- All modules **MUST** operate in isolated execution contexts.
- No module **MAY** access another module's internal state directly.
- Cross-module communication **MUST** occur only through explicit, documented interfaces.
- Global mutable state **MUST NOT** exist.
 - *"Global mutable state is prohibited."*

Deterministic Execution Requirements

- All **VNAGY** components **MUST** behave deterministically under identical inputs.
- Randomization **MUST NOT** be used in any security-relevant logic.
- Dynamic execution **MUST NOT** be used.
 - *"Dynamic code execution is prohibited."*
- All IO operations **MUST** be explicit and documented.
 - *"All IO operations must be explicit."*

Offline-First Requirements

- All detection, correlation, scoring, and response logic **MUST** execute locally.
- No VNAGY module **MAY** depend on cloud connectivity for security decisions.
- Local baselines, thresholds, and scoring profiles **MUST** be stored offline.
- Remote services **MAY NOT** influence operational behavior.

Supply-Chain Security Requirements

VNAGY includes a full supply-chain security layer (confirmed in your project structure):

"config-integrity-shield", "dependency-freeze", "digital-signatures", "hash-verification", "ingestion-anti-corruption", "offline-build-pipeline", "usb-guard"

Therefore:

- All dependencies **MUST** be frozen and cryptographically verified.
- All modules **MUST** include integrity markers validated at load time.
- All ingestion pipelines **MUST** include anti-corruption guards.
- USB or external ingestion **MUST** be sandboxed and validated.

Non-Reconstructability Requirements

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Your Coding Standard explicitly states:

"No operational logic is exposed." "Public documents contain exclusively symbolic constructs." "Any component that can be misused or reverse-engineered is not publicly released."

Therefore:

- Public VNAGY documents **MUST NOT** contain operational logic.
- Thresholds, weights, scoring parameters, correlation rules, and runtime behavior **MUST NOT** be published.
- Symbolic descriptions **MUST** be designed so they cannot be transformed into operational behavior.
- Architecture layers **MUST** remain strictly separated (conceptual vs operational).

Input Validation Requirements

- All inputs **MUST** be validated structurally and semantically.
- Invalid inputs **MUST** trigger explicit errors.
- Silent failure **MUST NOT** occur.
- Input normalization **MUST** be deterministic and reproducible.

Error Security Requirements

- All errors **MUST** be explicit and deterministic.
- Error messages **MUST NOT** leak internal logic, thresholds, or sensitive state.
- Error recovery paths **MUST** be documented and predictable.
- Catch-all errors **MUST NOT** exist.

Logging Requirements

Your Coding Standard states:

"Only structural events may be logged: start, stop, fail, recover."

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Therefore:

- Logs **MUST** contain only structural events.
- Logs **MUST NOT** contain sensitive data, heuristics, or detection logic.
- Debug logging **MUST NOT** exist.
- Logging **MUST** be minimal and deterministic.

Testing Requirements

- All tests **MUST** be deterministic.
- Randomization **MUST NOT** be used.
- Tests **MUST** run in isolation.
- Every function **MUST** have a unit test.
- Every module **MUST** have an integration test.
 - *"Tests must produce consistent results."*

VNAGY Code Examples

Module Structure Example (main.vng)

```
module main
```

```
import state
import errors
import types
```

```
function initialize(input)
  validate input
  state.transition("INIT")
  return types.Result("OK")
end
```

State Definition Example (state.vng)

```
state INIT
state READY
state FAILED
```

```
transition INIT → READY
transition READY → FAILED
transition FAILED → READY
```

Error Definition Example (errors.vng)

```
error InvalidStateError
  condition: state not in allowed set
  severity: high
  recovery: reset state to INIT
end
```

```
error MissingInputError
  condition: input is null
  severity: medium
  recovery: abort
end
```

Type Definition Example (types.vng)

```
type Result
  field status: string
end
```

```
type DeviceState
  field id: string
  field mode: string
end
```

Deterministic Function Example

```
function validate(input)
  if input is null
    raise MissingInputError
  end
end
```

Configuration Example (config.vng)

```
config
  DEFAULT_MODE = "offline"
  MAX_RETRY = 3
  TIMEOUT_MS = 5000
end
```

Component README Example (README .md)

device-detection

Purpose:

Deterministic offline-first device identification.

Inputs:

device_id

profile

Outputs:

DeviceState

Constraints:

No network access.

No dynamic execution.

No global state.

Appendix

Appendix sekcije **MAY** contain supplemental, non-normative material that supports the **VNAGY** standard. Appendix content **MUST NOT** introduce new requirements, rules, or constraints.

A. Terminology Table

Term	Definition
<u>VNAGY</u>	Offline-first security architecture designed for deterministic, isolated, and non-reconstructable operation.
<u>Module</u>	Smallest functional unit in VNAGY. Operates in isolation, has explicit inputs/outputs, and no global state.
<u>Component</u>	Logical grouping of modules forming a higher-level capability (e.g., agent, installer).
<u>.vng File</u>	Deterministic VNAGY module file containing symbolic structure definitions (types, states, errors).
<u>State</u>	Explicit, deterministic representation of module condition. No implicit transitions allowed.
<u>Transition</u>	Allowed movement between states. MUST be explicitly defined.
<u>Error</u>	Deterministic failure condition with explicit severity and recovery path.
<u>Type</u>	Structured data definition used by modules. MUST use PascalCase.
<u>Determinism</u>	Guarantee that identical inputs always produce identical outputs. No randomness, no dynamic execution.
<u>Offline-First</u>	VNAGY principle requiring all logic to run locally without cloud dependency.
<u>Isolation</u>	Modules cannot access each other's internal state; communication only through explicit interfaces.
<u>Explicit IO</u>	All input/output operations MUST be declared and documented. No hidden IO.

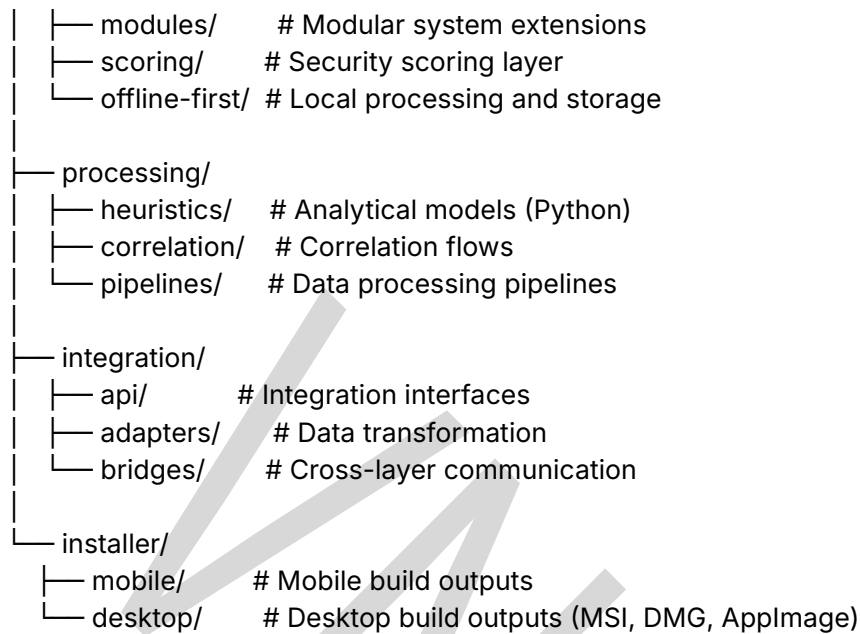
Term	Definition
<u>Non-Reconstructability</u>	Public documents contain symbolic constructs only; operational logic is never exposed.
<u>Integrity Marker</u>	Cryptographic signature or hash ensuring module authenticity.
<u>Dependency Freeze</u>	Supply-chain rule requiring all dependencies to be locked and verified.
<u>Config Parameter</u>	Deterministic configuration value with explicit type, default, and allowed range.
<u>Profile</u>	Offline behavioral template used by modules (e.g., detection profile, scoring profile).
<u>Sandbox</u>	Isolated execution environment preventing external influence or contamination.
<u>Structural Event</u>	Minimal logging event: start, stop, fail, recover. No operational details allowed.
<u>Security Boundary</u>	Explicit separation between conceptual and operational layers.

B. Directory Maps

```

vnagy/
├── ui/
│   ├── flutter/    # Mobile UI (Android & iOS)
│   ├── kotlin/     # Mobile service layer
│   ├── tauri/      # Desktop UI (Windows, Linux, macOS)
│   │   ├── webui/  # Desktop frontend (local rendering)
│   │   └── src-tauri/ # Desktop backend (Rust)
├── agent/
│   ├── config/      # Agent configuration
│   ├── integration/ # OS-level integration points
│   ├── src/         # Agent runtime logic
│   └── tests/       # Agent testing suite
├── agent-installer/
│   ├── config-generator/ # Installer configuration builder
│   ├── device-detection/ # Platform and hardware detection
│   ├── module-selection/ # Module selection logic
│   └── prof/         # Profiling and environment checks
├── core/
│   └── engine/      # Rust core engine

```



C. Example Structures

Non-operational symbolic examples:

- module structure
- state definitions
- error definitions
- type definitions
- config blocks

D. Document Version History

Version	Date	Author	Description
V1.0	20 July 2026	V. Nađ	Draft
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V1.2	01 August 2026	V. Nađ	Structural & Content Expansion Update

E. Future Extensions (Non-Normative)

This section **MAY** list ideas for future **VNAGY** expansions, npr.:

- additional module types
- extended offline-first profiles
- new deterministic patterns
- expanded documentation templates

F. References (Non-Normative)

Any external materials that inspired or informed the architecture, npr.:

- ISO/IEC standards
- NIST SP 800 series
- RFC documents
- academic papers

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IMAGY

1. Purpose

This document defines the **VNAGY** programming standard. The objective is to ensure minimal, deterministic, explicit, and structurally predictable code across all **VNAGY** modules and components.

VNAGY

2. Core Principles

2.1. Minimalism

Code must contain only what is necessary. No redundant constructs, no decorative abstractions, no “cleverness”.

2.2. Determinism

Execution must be predictable. No hidden side-effects, no implicit state transitions.

2.3. Explicitness

All behavior must be explicit. No magic values, no implicit conversions, no silent fallbacks.

2.4. Isolation

Modules must remain isolated. Cross-module dependencies are prohibited unless strictly required.

2.5. Predictable Flow

Control flow must remain linear and readable. Maximum nesting depth: 2 levels.

3. Naming Conventions

3.1. General

- English only
- Short, precise, descriptive
- No unnecessary abbreviations

3.2. Variables

``lower_snake_case``

3.3. Constants

``UPPER_CASE``

3.4. Functions

``verb_noun``

3.5. Classes / Types

``PascalCase``

3.6. Modules

``lower_snake_case``

4. Error Handling

4.1 Explicit Errors

All errors must be raised explicitly. Silent failure is prohibited.

4.2. Fail-Fast

Errors must be thrown immediately upon detection.

4.3. Predictable Format

Error structures must remain consistent across modules.

5. Logging Standard

5.1. Minimal Logging

Only structural events may be logged:

- start
- stop
- fail
- recover

5.2. No Verbose Output

Debug spam is prohibited.

6. Dependency Rules

6.1. Minimal Dependencies

External libraries must be avoided unless industry-standard.

6.2. Prefer Standard Library

Standard library usage is preferred whenever possible.

6.3. No Utility Abstractions

Libraries that “do everything” are prohibited.

7. Security Rules

7.1. No Global State

Global mutable state is prohibited.

7.2. No Reflection

Reflection is prohibited unless strictly required.

7.3. No Dynamic Execution

Dynamic code execution is prohibited.

7.4. No Implicit IO

All IO operations must be explicit.

8. Testing Standard

8.1. Deterministic Tests

Tests must produce consistent results.

8.2. No Randomization

Randomized behavior is prohibited.

8.3. Isolation

Tests must not depend on external state.

8.4. Coverage

Every function must have a unit test. Every module must have an integration test.

9. Documentation Standard

9.1. Minimal Documentation

Only essential information is documented.

9.2. Technical Tone

No marketing language. No storytelling.

9.3. Structure

- One paragraph per module
- One sentence per function

10. Code Review Rules

10.1. Structural Feedback Only

Subjective comments are prohibited.

10.2. Reject Complexity

Any unnecessary complexity must be removed.

10.3. Focus Areas

- correctness
- clarity
- security

11. VNAGY Architecture Rules

Implementation Exists; Public Access Limited to Non-Operational Content

11.1. Implementation Status

All VNAGY modules (M-Module, C-Pack, L-Pack, S-Pack, X-Pack) **exist in implemented form** within the private VNAGY development environment. The architecture is functional, modular, and internally validated.

11.2. Public Availability

Only **non-operational**, **non-computational**, and **non-reconstructable** components are made publicly available. Public documents contain exclusively:

- symbolic constructs
- conceptual interaction rules
- structural descriptions
- non-algorithmic behavior models
- licensing and IP constraints

No operational logic is exposed.

11.3. Restricted Content

The following implementation details are **not** publicly released:

- detection logic
- thresholds, weights, scoring parameters
- correlation rules
- runtime behavior
- operational semantics
- executable modules

- internal pipelines
- any component that can be misused or reverse-engineered

11.4. Separation of Layers

The conceptual layer (public) and the operational layer (private) remain strictly separated. No leakage of implementation details is permitted.

11.5. Security Constraint

All public VNAGY documents are intentionally designed to be **non-reconstructable**, ensuring that no symbolic description can be transformed into operational behavior.

11.6. Reference to VNAGY Whitepaper

This coding standard aligns with the architectural principles defined in: *VNAGY Secure-First Architecture — Internal Technical Specification v1.0 (Osijek / Bucharest)*. The whitepaper provides the conceptual and symbolic foundation of the system. The operational implementation exists but is not publicly released.

12. Intellectual Property & Licensing

12.1. Ownership

The VNAGY Secure-First Architecture, including all symbolic constructs, interaction rules, structural definitions, and conceptual models, is the exclusive intellectual property of the author.

All rights are reserved.

12.2. Permitted Use

The following actions are permitted:

- reading and personal study
- academic reference
- citation with attribution
- sharing the document in its original, unmodified form

12.3. Prohibited Use

The following actions are strictly prohibited:

- implementation of any construct, rule, or model
- derivation of operational logic from symbolic definitions
- reconstruction of algorithms, thresholds, weights, or detection semantics
- commercial use in any form
- modification, removal, or alteration of licensing terms
- redistribution of modified versions

12.4. Non-Operational Declaration

This document contains no operational details, no thresholds, no weights, no heuristics, no correlation logic, and no algorithmic semantics. All constructs are

symbolic and conceptual. The model is non-computational and non-reconstructable by design.

12.5. License

VNAGY Extended License (PSDL-1.3)

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VNAGY